

Name: _____

Date: _____

MULTIPLICATION, (6, 7, 9, 11 AND 12 TIMES TABLES)

LO: I can recall the harder times tables (6, 7, 9, 11, 12) and use them to solve problems.

Tricks for the harder tables

The **9 times table** trick: the digits of every multiple of 9 add to **9** (e.g. $36 \quad 3+6=9$). Or use $10 \times n$ then subtract n : $9 \times 7 = 70 - 7 = 63$.

11 times table trick: for 1-9, just repeat the digit ($4 \times 11 = 44$, $7 \times 11 = 77$).

Patterns to spot

●	$9 \times 7 = 63$ ($6+3=9$)	9-trick
●	$11 \times 5 = 55$	doubled digit
●	$12 \times 6 = 72$	double 6 = 12, $\times 6 = 72$
●	$7 \times 8 = 54$	should be 56
●	$11 \times 12 = 121$	should be 132

Key idea

Use friendly facts to bridge to harder ones. e.g. $7 \times 6 = (7 \times 5) + 7 = 35 + 7 = 42$.

$$\boxed{9 \times 8} = \boxed{10 \times 8} - 8 = \boxed{72}$$

$\times 10$ then subtract

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - 9 times table trick

Calculate 9×6

$$10 \times 6 = 60$$

Subtract one 6

$$60 - 6 = 54$$

$$= 54$$

Multiples of 9 are always one less than $10 \times n$.

Example 2 - 7 times table

Calculate 7×8

$$7 \times 7 = 49$$

Add one more 7

$$49 + 7 = 56$$

$$= 56$$

Use the square fact $7 \times 7 = 49$ as a stepping stone.

Example 3 - 12 times table

Calculate 12×8

$$12 = 10 + 2$$

$$(10 \times 8) + (2 \times 8)$$

$$80 + 16 = 96$$

$$= 96$$

Split 12 into $10 + 2$ and multiply each separately.

Example 4 - 11 times table

Calculate 11×7

For 1-9, repeat the digit

$$11 \times 7 = 77$$

$$= 77$$

11×12 breaks the rule: it's 132, not 1212.

